

GATE SOLUTION IN MINING ENGINEERING

(Containing Solved Questions 2015 - 2007)

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Gate Solution-2013

- Q.1** In the Coward flammability diagram, the respective percentages of methane and oxygen at the nose limit are
- (A) 14.2, 0.0 (B) 14.1, 18.2 (C) 5.8, 12.1 (D) 5.0, 19.2

Answer (C)

- Q.2** If the transpose of a matrix is equal to its inverse, then the matrix is
- (A) symmetric (B) orthogonal (C) skew symmetric (D) singular

Answer (B)

Solution .

As per given condition-

$$\Rightarrow A' = A^{-1}$$

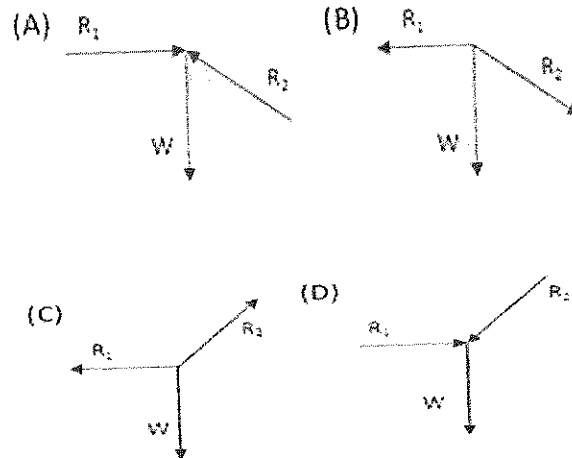
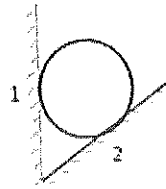
$$\Rightarrow A'A = 1$$

Hence , matrix is orthogonal.

- Q.3** In the Moh's scale of hardness, the minerals in increasing sequence of hardness are
- (A) calcite, gypsum, topaz, diamond (B) topaz, gypsum, calcite, diamond
- (C) calcite, gypsum, diamond, topaz (D) gypsum, calcite, topaz, diamond

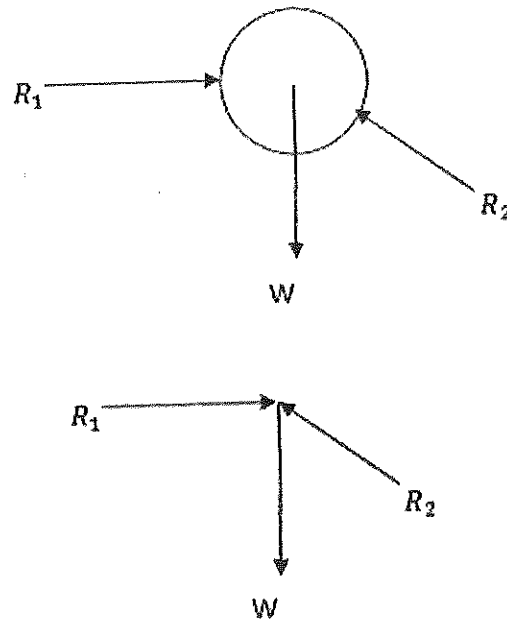
Answer (D)

- Q.4** A ball of weight W is supported on smooth walls as shown in the following figure. R_1 and R_2 are reactions from the walls 1 and 2. The free body diagram of the ball is represented by



Answer (A)

Solution .



Q.5 For a 25 mm diameter spherical charge, the maximum allowable charge length in cm is

- (A) 15.0 (B) 25.0 (C) 30.0 (D) 150.0

Answer (A)

Solution .

Explosive charges used in Vertical Crater Retreat method are normally spherical.

Spherical charges have been defined as having a length : diameter (L:D) ratio of 6:1.

∴ diameter of spherical charges = 25 mm

∴ length of spherical charges = $6 \times 25 = 150 \text{ mm} = 15 \text{ cm}$

Q.6 Long-hole drilling with crater blasting is used for the construction of

- (A) winze (B) shaft (C) raise (D) decline

Answer (C)

Q.7 Rill stoping method is a form of

- (A) block caving
(B) artificially supported stoping
(C) underhand stoping
(D) overhand stoping

Answer (D)

Q.8 Transit theodolite is a

- (A) micro-optic theodolite
(B) theodolite with face left and face right reading facilities

(C) theodolite with stadia hairs

(D) theodolite with two vertical circles

Answer (B)

Q.9 Incubation period is NOT related to

(A) crossing point temperature of coal

(B) panel size

(C) seam thickness

(D) explosibility of coal dust

Answer (D)

Q.10 The rotational speed and cutting velocity of a drill are 350 rpm and 71.50 m/min respectively. The diameter of the rotary drill bit in mm is

(A) 65

(B) 67

(C) 68

(D) 70

Answer (A)

Solution.

Radial speed is given by-

$$v = 2\pi\omega r$$

Where,

ω = rotational speed = 350 rpm

r = radial distance

$$\therefore 71.50 = 2 \times 3.14 \times 350 \times r$$

$$\therefore r = 0.0325 \text{ m} = 32.5 \text{ mm}$$

$$\therefore \text{diameter} = 2 \times 0.0325 \text{ m} = 65 \text{ mm}$$

- Q.11** The pressure on a phreatic surface is
- (A) less than atmospheric pressure
 - (B) greater than atmospheric pressure
 - (C) equal to atmospheric pressure
 - (D) independent of atmospheric pressure

Answer (C)

- Q.12** Events A and B are independent but NOT mutually exclusive. If the probabilities $P(A)$ and $P(B)$ are 0.5 and 0.4 respectively, then $P(A \cup B)$ is
- (A) 0.6 (B) 0.7 (C) 0.8 (D) 0.9

Answer (B)

Solution .

$$P(A) = 0.5$$

$$P(B) = 0.4$$

$\therefore$ Events A & B NOT mutually exclusive, then

$$P(A \cap B) \neq 0$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\text{Or } P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$$

$$P(A \cup B) = 0.5 + 0.4 - 0.5 \times 0.4$$

$$\therefore P(A \cup B) = 0.7$$

- Q.13** Among the following options, the specific energy for rock-drilling is lowest in

- (A) rotary diamond drilling (B) rotary roller drilling
(C) percussive drilling (D) jet piercing

Answer (C)

Q.14 Identify the correct statement for a 'normal distribution'.

- (A) Mean is greater than mode but less than median
(B) Mean is less than mode but greater than median
(C) Mean is greater than mode and median
(D) Mean, median and mode are equal

Answer (D)

Q.15 An emulsion explosive of specific gravity 1.25 is used for blasting in an iron ore formation having P-wave velocity of 3000 m/s and specific gravity of 3.20. For an explosive impedance to rock impedance ratio of 0.5, the desired velocity of detonation of the explosive in m/s is

- (A) 3840 (B) 4000 (C) 4200 (D) 7680

Answer (A)

Solution .

$$\therefore \frac{\text{explosive impedance}}{\text{rock impedance}} = 0.5$$

$$\Rightarrow \frac{(VOD)_e \times \text{specific gravity of explosive}}{(VOD)_r \times \text{specific gravity of iron ore}} = 0.5$$

$$\Rightarrow \frac{(VOD)_e \times 1.25}{3000 \times 3.20} = 0.5$$

$$\Rightarrow (VOD)_e = \frac{0.5 \times 3000 \times 3.20}{1.25}$$

$$\Rightarrow (VOD)_e = 3840 \text{ m/s}$$

The number of ways in which the letters in the word MINING can be arranged is

(A) 90

(B) 180

(C) 360

(D) 720

Answer (B)

MINING = 6

N = 2 & I = 2

$$\therefore \text{Required number of ways} = \frac{6!}{2!2!} = \frac{720}{4} = 180$$

Under standard temperature and pressure conditions the theoretical maximum height in m to which water can be lifted using an air-lift pump is

(A) 10.33

(B) 9.61

(C) 7.45

(D) 6.05

Answer (A)

In a belt conveyor system, function of the snub pulley is to

(A) clean the inner surface of the belt

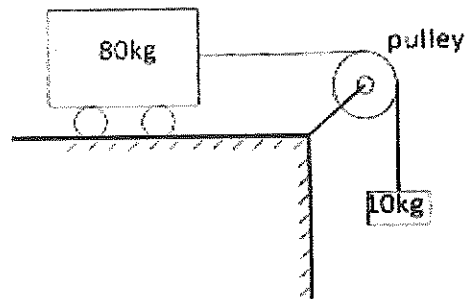
(B) clean the outer surface of the belt

(C) increase the angle of contact of belt with drive drum

(D) decrease the belt tension

Answer (C)

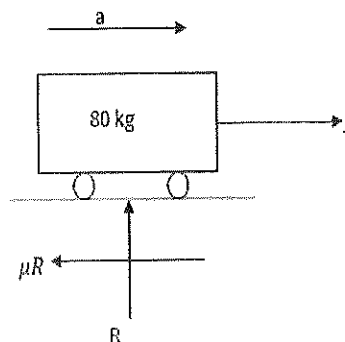
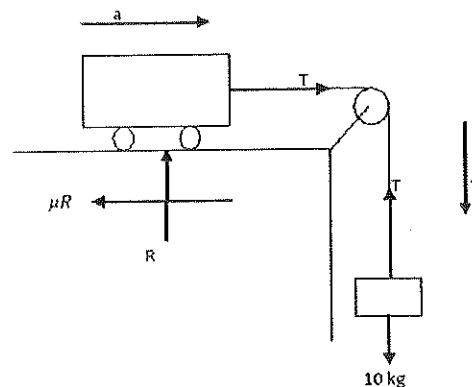
- Q.19 In the following figure, the coefficient of kinetic friction between the trolley and the surface is 0.04. When the block is released from rest, the acceleration of the trolley in m/s^2 becomes



- (A) 9.65 (B) 1.23 (C) 1.09 (D) 0.74

Answer (D).

Solution .



$$\sum V = 0$$

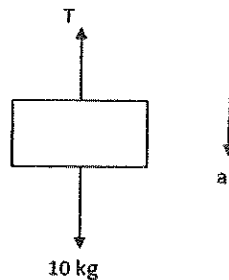
$$R - 80 = 0$$

$$R = 80 \times 9.81 \text{ N}$$

Applying $F = ma$

$$T - \mu R = 80 a$$

$$T - 0.04 \times 80 \times 9.81 = 80 a \dots\dots\dots(1)$$



again applying $F = ma$,

$$10 \times 9.81 - T = 10 a \dots\dots\dots(2)$$

Solving equation (1) & (2)

$$\therefore a = 0.74 \text{ m/sec}$$

Q.20 Two meshing spur gear wheels of Module 6 have 24 and 42 teeth. The distance in mm between the centres of the gear wheels is

- (A) 1000 (B) 198 (C) 126 (D) 72

Answer (B)

Solution.

We have,

$$\text{Module} = \frac{\text{diameter}}{\text{teeth}}$$

⇒ For first meshing spur gear wheel

$$6 = \frac{(\text{diameter})_1}{24}$$

$$\therefore (\text{diameter})_1 = 6 \times 24 = 144 \text{ mm}$$

$$\text{Or } (\text{radius})_1 = 72 \text{ mm}$$

⇒ For second meshing spur gear wheel

$$6 = \frac{(\text{diameter})_2}{42}$$

$$\therefore (\text{diameter})_2 = 6 \times 42 = 252 \text{ mm}$$

$$\text{Or } (\text{radius})_2 = 126 \text{ mm}$$

Therefore, distance between the centers of gear wheels = $72 + 126 = 198 \text{ mm}$

- Q.21** In an experiment to study coal dust explosibility, it is found that at least 3.0 g of limestone dust should be added to a sample of 2.0 g of coal dust to ensure that propagation of flame does not take place. The explosibility factor of coal dust is
- (A) 60.00 (B) 20.00 (C) 6.70 (D) 1.50

Answer (D)

Solution .

Explosibility factor is the ratio between inert and combustible content.

Therefore,

$$\text{Explosibility factor} = \frac{3}{2} = 1.5$$

- Q.22** A 20 m steel tape used in a mine survey is found to be 20 cm short when compared with a standard tape. If the measured volume of a dump using the tape is 4000 m^3 , its actual volume in m^3 is

(A) 3881

(B) 3902

(C) 3920

(D) 4121

Answer (A)

Solution.

We have,

$$\text{Actual volume} = \text{incorrect volume} \times \left(\frac{L'}{L}\right)^3$$

$$\text{Where, } L' = \text{incorrect length of chain} = 20 - 0.2 = 19.8 \text{ m}$$

$$L = \text{correct length of chain} = 20 \text{ m}$$

$$\text{Now, actual volume} = 4000 \times \left(\frac{19.8}{20}\right)^3 = 3881 \text{ m}^3$$

Q.23 A mine worker inhales normal air; whereas, the exhaled air contains 16.65% O_2 and 3.83% CO_2 . The respiratory quotient of breathing for the worker is

(A) 0.23

(B) 0.89

(C) 0.99

(D) 1.13

Answer (B)

Solution .

$$\begin{aligned}\text{Respiratory quotient} &= \frac{\text{exhaled } CO_2}{O_2 \text{ consumed}} \\ &= \frac{3.83}{20.93 - 16.65} = 0.89\end{aligned}$$

Q.24 Block economic values in Lakhs of Rupees for a section of a block economic model are shown below.

-1	-1	1	-1	0	-1
-1	0	0	0	-1	-2
-5	-3	-2	5	-2	-3

At a permissible slope angle of 1:1, the optimum pit value of the section in Lakhs of Rupees is

- (A) 0 (B) 1 (C) 2 (D) 3

Answer (C)

Solution .

The following steps are used-

- (1) The cone is floated from left to right along the top row of the blocks in the section, if there is a positive block it is removed.

-1	-1		-1	0	-1	+1
-1	0	0	0	-1	-2	
-5	-3	-2	5	-2	-3	

- (2) After traversing the first row the apex of the cone is moved to the second row, starting floats from left to right stopping when it encounters the first positive blocks. If the sum of all blocks falling within the cone is positive (or zero), these blocks are removed . if the sum is negative the blocks are left and the cone floats to the next positive blocks on this row. The summing and mining or leaving process is repeated.

-1						-1-1+0-1-1+0+0+5=+1
-1	0				-2	
-5	-3	-2		-2	-3	

- (3) This floating cone process moving from left to right and top to bottom of the section continues until no more blocks can be removed.

- (4) The profitability of the section is found by summing the value of the blocks removed.

$$\therefore \text{The optimum pit value of section} = +1 + 1 = +2$$

$$= 2 \text{ lakh of rupees}$$

Optimum pit limit
↓

-1	-1	1	-1	0	-1
-1	0	0	0	-1	-2
-5	-3	-2	5	-2	-3

Q.25 The boundary of a mine is plotted on a scale of 1:2000. If a planimeter measures the plotted area as 58 cm^2 , the actual mine area in m^2 is

- (A) 5800 (B) 11600 (C) 23200 (D) 29000

Answer (C).

Solution.

$$R.F = \frac{\text{Dimension of object in drawing}}{\text{actual dimension of object}}$$

Given , plotted area = 58 cm^2

So, plotted length or width = $\sqrt{58} \text{ cm}$

$$\therefore \frac{1}{2000} = \frac{\sqrt{58} \text{ cm}}{\text{actual length or width of object}}$$

$\therefore$ actual length or width of object = $2000 \times \sqrt{58} \text{ cm}$

So, actual area of object = $232000000 \text{ cm}^2 = 23200 \text{ m}^2$

Q.26 For a shrinkage stope the following data values are given

Insitu tonnage	:	9000 tonne
Insitu grade	:	5.2 g/tonne
Average grade of waste	:	1.4 g/tonne

Loss of ore in the stope . 10%

Dilution : 20%

The grade at the mill-head in g/tonne is _____

Answer : 4.5 to 4.518.

Solution .

	Tons	Grade (g/ton)	TonsXGrade (gram)
Ore in place	9000	5.2	9000×5.2=46800
Dilution (20 %)	$(+)\frac{20}{100} \times 9000 = (+)1800$	1.4	$(+) 1800 \times 1.4 = (+) 2520$
Ore loss (10%)	$(-)\frac{10}{100} \times 9000 = (-)900$	5.2	$(-)900 \times 5.2 = (-) 4680$
Total	9900		44640

$$\therefore \text{Mill grade} = \frac{44640}{9900} \text{ g/m} = 4.5 \text{ g/tonne}$$

Q.27 In an experiment to determine specific gravity of a soil sample, the following data is obtained:

Mass of empty pycnometer : 20.4 g

Mass of pycnometer with soil sample : 51.6 g

Mass of pycnometer with soil sample filled with water : 88.6 g

Mass of pycnometer filled with water : 70.4 g

The specific gravity of the sample is _____

Answer : 2.35 to 2.45

Solution .

$$\text{Specific gravity} = \frac{W_1}{W_1 - W_2}$$

Where, W_1 = weight of solid in air = $51.6 - 20.4 = 31.2$ g

W_2 = weight of solid in water = $88.6 - 70.4 = 18.2$ g

$$\therefore \text{Specific gravity of sample} = \frac{31.2}{31.2 - 18.2} = 2.40$$

A cylindrical rock specimen of diameter 54 mm has Young's modulus of 68.97 GPa and Poisson's ratio of 0.35. The rock specimen fails in uniaxial compression at a lateral strain of 0.01%. The axial load at failure in kN is _____

Answer : 44.8 to 45.5

$$\text{Poisson ratio} = \frac{\text{lateral strain}}{\text{longitudinal strain}}$$

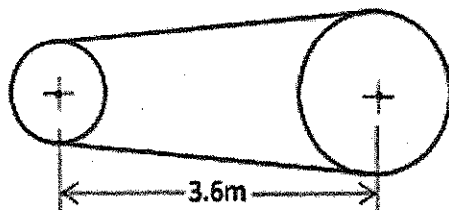
$$0.35 = \frac{0.01\%}{\varepsilon} = \frac{10^{-4}}{P/AE} \quad [\text{from Hook's law, } \varepsilon = P/AE]$$

$$\therefore P = \frac{10^{-4}}{0.35} \times 3.14 \times 27^2 \times 10^{-6} \times 68.97 \times 10^9$$

$$\therefore P = 45.107 \text{ kN}$$

An open belt drive connects two pulleys on parallel shafts that are 3.6 m apart as shown in the figure. The diameters of the pulleys are 2.4 m and 1.6 m. The angle of contact on the smaller pulley in degrees is _____

Answer : 166 to 169



Solution .

$$\sin \theta = \frac{r_1 - r_2}{d}$$

Where,

$$r_1 = \text{radius of larger pulley} = \frac{2.4}{2} = 1.2 \text{ m}$$

$$r_2 = \text{radius of smaller pulley} = \frac{1.6}{2} = 0.8 \text{ m}$$

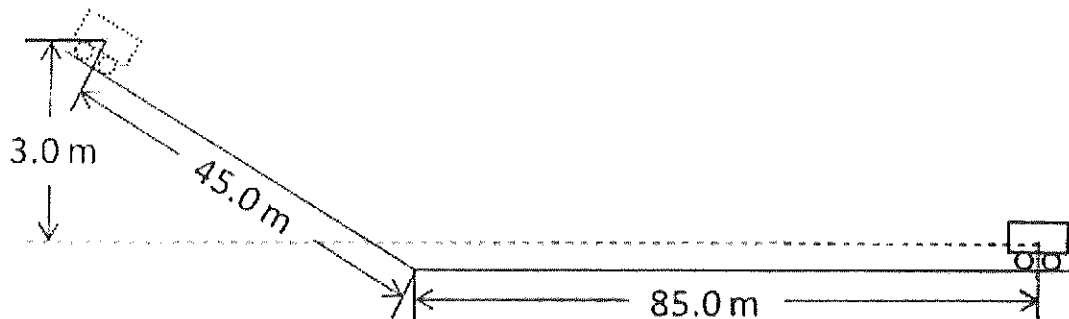
D = distance between centres of both pulley = 3.6 m

$$\therefore \sin \theta = \frac{1.2 - 0.8}{3.6} = 0.11$$

$$\therefore \theta = 6.3^\circ$$

$$\begin{aligned} \therefore \text{Angle of contact for smaller pulley} &= 180^\circ - 2 \times \theta \\ &= 180^\circ - 2 \times 6.3^\circ \\ &= 167^\circ 24' 0'' \end{aligned}$$

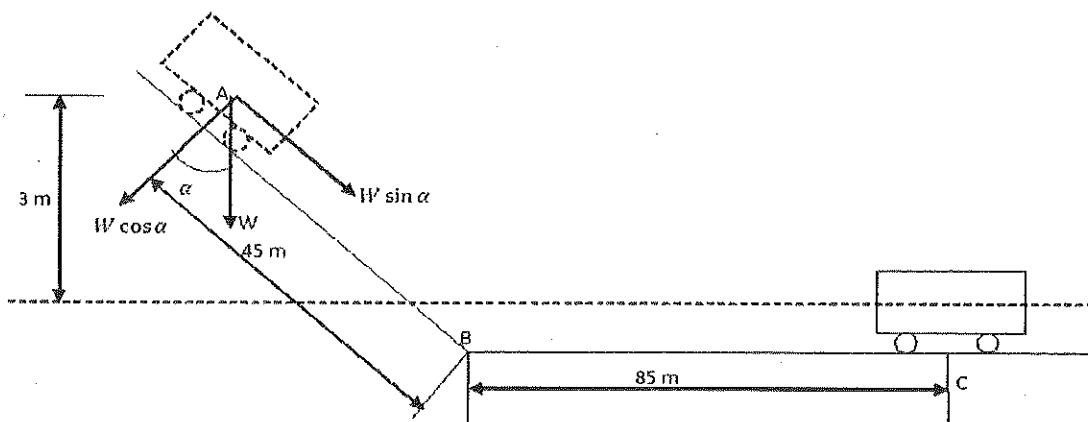
- Q.30** A two tonne mine car is released from the top of an incline at a height of 3 m as shown in the figure. The mine car travels 45 m along the inclined track and another 85 m along the horizontal track before coming to rest.



The specific rolling resistance of the car in N/tonne is _____

Answer : 225 to 231.

Solution .



$$W = mg = 2000 \times 9.81 = 19620 \text{ N}$$

$$\alpha = \sin^{-1}\left(\frac{3}{45}\right) = 3.8^\circ$$

$$\text{Rolling resistance} = \mu R$$

Here,

Potential energy = (rolling resistance from A to B) × (distance from A to B) + (rolling resistance from B to C) × (distance from B to C)

$$m.g.h = \mu mg \cos 3.8^\circ \times 45 + \mu mg \times 85$$

$$19620 \times 3 = \mu mg [\cos 3.8^\circ \times 45 + 85]$$

$$\text{Or Rolling resistance} = \mu mg = \frac{19620 \times 3}{\cos 3.8^\circ \times 45 + 85}$$

$$\text{Rolling resistance} = 453.11 \text{ N/2 tonne}$$

$$\text{Rolling resistance} = 226.6 \text{ N/tonne}$$

Q.31 A surface miner with 2.0 m cutting drum width excavates coal in windrowing

mode from a bench with effective face length 200 m. The cutting speed of the surface miner is 10 m/min and the cutting depth 25 cm. The density of coal is 1.4 tonne/m³. If the average turning time of the machine at the face end is 5 min, the rate of production in tonne/hour becomes _____

Answer : 335 to 337.

Solution.

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

$$\text{speed} = \frac{\text{face length}}{\text{time}}$$

$$\text{time} = \frac{200}{10} = 20 \text{ minutes}$$

Now, total time taken in one cut = 20 + 5 = 25 minutes

$$\text{production by shovel} = 2 \times 200 \times 0.25 = 100 \text{ m}^3$$

$$= 100 \times 1.4 = 140 \text{ tonne}$$

Now, In 25 minutes = 140 tonnes

$$\therefore \text{In 1 hour} = \frac{140 \times 60}{25} = 336 \text{ tonnes}$$

Hence, rate of production in tonne/hour = 336 tonne /hour

Q.32 A core sample of a rock, having diameter 54 mm and length 108 mm, is subjected to axial loading. If the axial strain and Poisson's ratio are 2000×10^{-6} and 0.28 respectively, the value of volumetric strain, represented in micro-strain is _____

Answer : 0.000875 to 0.000885.

Solution.

$$\text{Volumetric strain} = \frac{\Delta V}{V} = \epsilon (1 - 2\theta)$$

Where, ϵ = axial strain = 2000×10^{-6}

ν = poisson ratio = 0.28

$\therefore$ Volumetric strain = $2000 \times 10^{-6} (1 - 2 \times 0.28)$

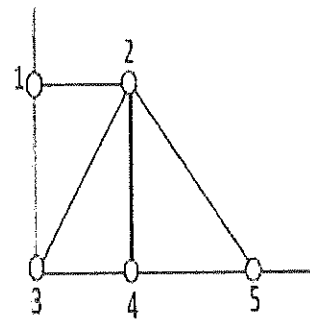
$\therefore$ Volumetric strain = 0.00088

Q.33

A flat bauxite deposit has thickness of 10 m with an average density of 2200 kg/m³. The grade values and the sample coordinates are as shown in the table.

To carry out reserve estimation using triangular method, the triangles are constructed as shown in the figure.

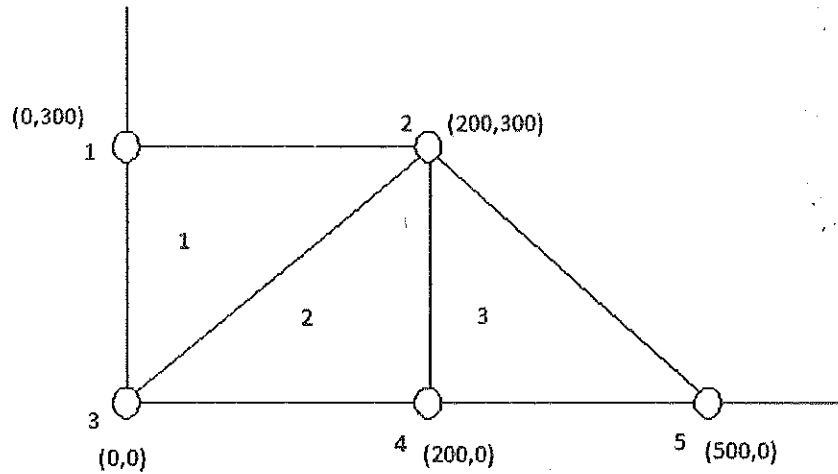
Sample No	1	2	3	4	5
Alumina%	35	40	39	47	42
x coordinate, m	0	200	0	200	500
y coordinate, m	300	300	0	0	0



The alumina content in million tonnes, in the region comprising the three triangles is

Answer : 0.9 to 1.

Solution.



$$g_{\Delta_1} = \frac{35 + 40 + 39}{3} = 38\%$$

$$g_{\Delta_2} = \frac{47 + 40 + 39}{3} = 42\%$$

$$g_{\Delta_3} = \frac{47 + 40 + 42}{3} = 43\%$$

And

$$A_1 = \frac{1}{2} \times 200 \times 300 = 30,000 \text{ m}^2$$

$$A_2 = \frac{1}{2} \times 200 \times 300 = 30,000 \text{ m}^2$$

$$A_3 = \frac{1}{2} \times 300 \times 300 = 45,000 \text{ m}^2$$

$$\text{Grade of deposit (g)} = \frac{A_1 g_{\Delta_1} + A_2 g_{\Delta_2} + A_3 g_{\Delta_3}}{A_1 + A_2 + A_3}$$

$$g = 41.29 \%$$

$$\text{Volume of deposit (V)} = (A_1 + A_2 + A_3) \times \text{thickness}$$

$$= (30000 + 30000 + 45000) \times 10$$

$$= 1050000 \text{ m}^3$$

$$\text{Tonnage of deposit} = \frac{\text{density } (\rho) \times V}{1000}, \text{ tonne}$$

$$= \frac{2200 \times 1050000}{1000}, \text{ tonne}$$

$$= 2310000 \text{ tonne}$$

$$= 2.31 \text{ million tonne}$$

$$\therefore \text{alumina content} = 2.31 \times 0.4129$$

$$= 0.95 \text{ million tonne}$$

Q.34 If the following linear system of equations has non-trivial solutions

$$px + y + z = 0$$

$$2x + y - 2z = 0$$

$$x + 2y - 3z = 0$$

the value of p is

(A) 1

(B) 0

(C) -1

(D) -7

Answer (D)

Solution.

In case of Non-trivial solution

$$\Delta = 0$$

$$\begin{vmatrix} P & 1 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & -3 \end{vmatrix} = 0$$

$$P \begin{vmatrix} 1 & -2 \\ 2 & -3 \end{vmatrix} - 1 \begin{vmatrix} 2 & -2 \\ 1 & -3 \end{vmatrix} + 1 \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 0$$

$$\therefore P + 4 + 3 = 0$$

$$\therefore P = -7$$

Q.35 A bucket wheel excavator with 20 buckets of capacity 0.5 m^3 each, rotates at 5 rev/min. The bucket fill factor is 80%. The excavator loads on to 1200 mm wide belt conveyor. The cross-section area (m^2) of the material on the belt is $0.1B^2$, where B is the belt width in m. The minimum speed of the belt in m/s

to avoid spillage of material is

(A) 7.23

(B) 5.79

(C) 4.63

(D) 3.70

Answer (C)

Solution.

Productivity of bucket wheel excavator is given by:

$$Q = 60 \times \text{wheel rpm} \times \text{No. of bucket} \times \text{bucket capacity, } m^3/h$$

$$Q = 60 \times 5 \times 20 \times 0.5 = 3000 \text{ } m^3/h = 0.84 \text{ } m^3/sec$$

$$\therefore Q = 0.84 \times 0.80 = 0.7 \text{ } m^3/sec$$

And productivity of belt conveyor is given by:

$$T = abv \text{ } m^3/sec$$

$$\therefore T = 0.1 \times 1.2^2 \times 1 \times v \text{ } m^3/sec$$

$$\text{And now, } T = Q$$

$$0.1 \times 1.2^2 \times 1 \times v = 0.7$$

$$\therefore v = 4.63 \text{ } m/sec$$

Q.36 A simplex table shown below is generated during the maximization of a linear programming problem using simplex method

Variable	Z	X1	X2	X3	X4	RHS
Z	1	-1	0	1	0	6
X2	0	1/3	1	1/3	0	2
X4	0	7/3	0	-2/3	1	2

After one iteration, the value of the objective function becomes

(A) 48/7

(B) 11/3

(C) 22/7

(D) 2/3

Answer (A)

Solution .

Iteration procedure-

Step 01. Select a most negative coefficient in row (0). Put a box around the column below this coefficient and call this the pivot column

Step 02. Determine the minimum ratio test-

- Pick out each coefficient in pivot column that is strictly positive (> 0)
- Divide each of these coefficient in to the right side for same row.
- Identify the same row that the smallest of these ratio.

variable	eq.	coefficient of					R.H.S	Ratio
		Z	X_1	X_2	X_3	X_4		
Z	0	1	-1	0	1	0	6	
X_2	1	0	1/3	1	1/3	0	2	$\frac{2 \times 3}{1} = 6$
X_4	2	0	7/3	0	-2/3	1	2	$\frac{2 \times 3}{7} = \frac{6}{7}$ (minimum) ←

- Put a box around this row and call it pivot and also call the number that is in both boxes the pivot number.

Z	0	1	-1	0	1	0	6
X_2	1	0	1/3	1	1/3	0	2
X_4	2	0	7/3	0	-2/3	1	2

Step 03. Divide the pivot row by pivot number and obtained row called new pivot row.

Step 04. Performing operation – Row (0) → Row (0) + Row (2)

Performing operation – Row (1) → Row (1) – $\frac{1}{3}$ Row (2)

Iteration variable	eq.	coefficient of					R,H.S	
		z	X_1	X_2	X_3	X_4		
0	z	0	1	-1	0	1	0	6
	X_2	1	0	1/3	1	1/3	0	2
	X_4	2	0	7/3	0	-2/3	1	2
1	z	0	1	0	0	15/21	3/7	48/7
	X_2	1	0	0	1	3/7	-3/21	12/7
	X_1	2	0	1	0	-6/21	3/7	6/7

After one iteration , the value of the objective function becomes = 48/7

Q.37 The value of $\int_0^{\pi} \log(\cos x) dx$ is

- (A) $-\pi/2 \log 2$ (B) $-\pi/4 \log 2$ (C) $\pi/2 \log 2$ (D) $\pi/4 \log 2$

Answer (A)

Solution.

$$\text{Let, } I = \int_0^{\pi/2} \log_e \cos x dx \dots\dots\dots(1)$$

$$I = \int_0^{\pi/2} \log_e \cos \left(\frac{\pi}{2} - x \right) dx$$

$$I = \int_0^{\pi/2} \log_e \sin x dx \dots\dots\dots(2)$$

$$[\because \text{properties } \int_0^a f(x) dx = \int_0^a f(a-x) dx]$$

Adding (1) & (2)

$$2I = \int_0^{\pi/2} \log_e \cos x \, dx + \int_0^{\pi/2} \log_e \sin x \, dx$$

$$2I = \int_0^{\pi/2} [\log_e \cos x + \log_e \sin x + \log_e 2 - \log_e 2] dx$$

$$2I = \int_0^{\pi/2} [\log_e (2 \sin x \cos x) - \log_e 2] dx$$

$$2I = \int_0^{\pi/2} \log_e \sin 2x \, dx - \int_0^{\pi/2} \log_e 2 \, dx$$

$$2I = \int_0^{\pi/2} \log_e \sin 2x \, dx - \frac{\pi}{2} \log_e 2$$

In first integral, putting $2x = \theta$

$$\text{Then } dx = \frac{d\theta}{2}$$

When $x = 0, \theta = 0$ and when $x = \frac{\pi}{2}, \theta = \pi$

$$\therefore 2I = \int_0^{\pi} \log_e \sin \theta \, d\theta - \frac{\pi}{2} \log_e 2$$

$$2I = 2 \cdot \frac{1}{2} \int_0^{\pi/2} \log_e \sin \theta \, d\theta - \frac{\pi}{2} \log_e 2$$

$$2I = I - \frac{\pi}{2} \log_e 2$$

$$\therefore I = -\frac{\pi}{2} \log_e 2$$

Q.38 Given the following,

Machine	Component
P. Dint header	1. Cowl
Q. Coal plough	2. Cutting chain
R. Road header	3. Loading apron
S. Shearer	4. Static set of bits

the correct match is

(A) P-4, Q-2, R-3, S-1 (B) P-3, Q-4, R-2, S-1

(C) P-2, Q-3, R-4, S-1 (D) P-2, Q-4, R-3, S-1

Answer (D)

Q.39

Given the following,

Rescue apparatus	Characteristic
P. Draeger BG-4	1. Open circuit chemical oxygen self-rescuer
Q. MSA IW-65	2. Filter type self-rescuer
R. Draeger Pulmotor	3. Self-contained breathing apparatus
S. Oxyboks	4. Resuscitation apparatus

the correct match is

- | | |
|-------------------------------|-------------------------------|
| (A) P-3, Q-2, R-1, S-4 | (B) P-4, Q-1, R-2, S-3 |
| (C) P-3, Q-2, R-4, S-1 | (D) P-1, Q-4, R-3, S-2 |

Answer (C)

Q.40

Given the following,

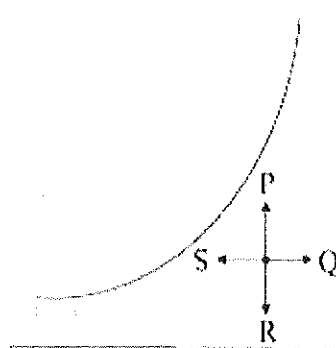
Equation/formula/law	Application
P. Bernoulli equation	1. Pressure loss in laminar flow of fluid
Q. Poiseuille equation	2. Drag loss due to regular obstructions in fluid flow
R. Bromilow's formula	3. Energy conservation in ideal fluid flow
S. Stokes law	4. Terminal settling velocity of fine particles in fluid

the correct match is

- | | |
|-------------------------------|-------------------------------|
| (A) P-3, Q-1, R-2, S-4 | (B) P-1, Q-3, R-2, S-4 |
| (C) P-2, Q-3, R-4, S-1 | (D) P-3, Q-1, R-4, S-2 |

Answer (A)

Q.41 Four psychrometric processes P, Q, R and S are shown in the psychrometric chart below.



These processes respectively represent

- (A) dehumidification, humidification, sensible heating, sensible cooling
- (B) sensible heating, humidification, dehumidification, sensible cooling
- (C) dehumidification, sensible heating, sensible cooling, humidification
- (D) humidification, sensible heating, dehumidification, sensible cooling

Answer (D)

Q.42 Given the following differential equation

$$\frac{d^2y}{dx^2} + 7\frac{dy}{dx} + 12y = 0$$

the general solution is

- | | |
|------------------------------|-------------------------------|
| (A) $y = Ae^{4x} + Be^{-3x}$ | (B) $y = Ae^{-4x} + Be^{-3x}$ |
| (C) $y = Ae^{3x} + Be^{-4x}$ | (D) $y = Ae^{4x} + Be^{3x}$ |

Answer (B)

Solution.

We have

$$\frac{d^2y}{dx^2} + 7 \frac{dy}{dx} + 12y = 0$$

In the symbolic form-

$$D^2 y + 7D y + 12y = 0$$

$$(D^2 + 7D + 12)y = 0$$

Its auxiliary equation will be -

$$D^2 + 7D + 12 = 0$$

$$\therefore D = -3, -4$$

Hence, general solution will be-

$$y = Ae^{-4x} + Be^{-3x}$$

Q.43 Given the following,

Mining method

Technique

P. Thick seam extraction

1. Double unit face

Q. Bord and pillar extraction

2. Jet cutting

R. Longwall face development

3. Inclined slicing

S. Hydraulic mining

4. Half moon method

the correct match is

(A) P-3, Q-4, R-2, S-1

(B) P-1, Q-4, R-2, S-3

(C) P-3, Q-2, R-1, S-4

(D) P-3, Q-4, R-1, S-2

Answer (D)

Q.44 Given the following,

Excavating/loading machine

Transportation scheme

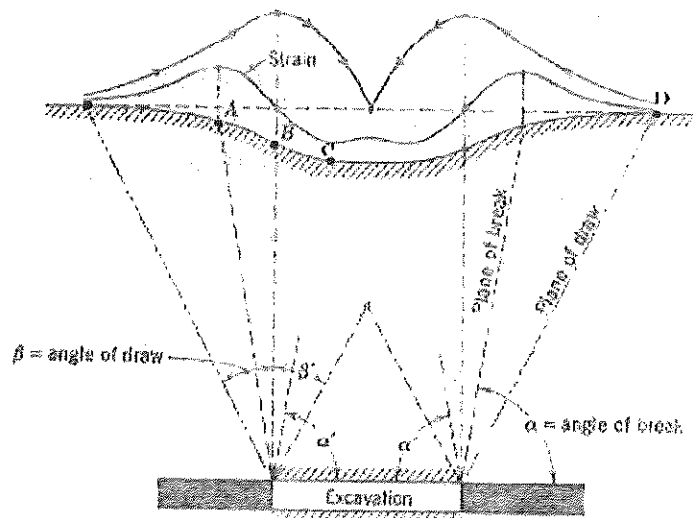
- | | |
|---------------------------|-------------------------------------|
| P. Bucket Wheel Excavator | 1. Mine tub |
| Q. Continuous Miner | 2. Armoured flexible chain conveyor |
| R. Shearer | 3. Shiftable Conveyor |
| S. Load Haul Dumper | 4. Shuttle car |

the correct match is

- (A) P-3, Q-2, R-4, S-1 (B) P-3, Q-4, R-2, S-1
 (C) P-3, Q-2, R-1, S-4 (D) P-1, Q-4, R-3, S-2

Answer (B)

- Q.45 A sub-critical subsidence profile is shown in the figure below. The points A, B, C, and D represent respectively the points of



- (A) zero vertical displacement, maximum tension, inflexion, maximum compression
 (B) inflexion, maximum tension, maximum compression, zero vertical displacement
 (C) maximum tension, inflexion, maximum compression, zero vertical displacement
 (D) maximum compression, maximum tension, inflexion, zero vertical displacement

Answer (C)

Q.46 The uniaxial compressive strength of a limestone sample is 80 MPa. The sample is confined at a pressure of 20 MPa in a triaxial compressive strength test. Based on Hoek-Brown failure criteria the maximum principal stress at failure in MPa is (consider rock constants as $m = 7.88$, $s = 1.0$ and $a = 0.5$)

- (A) 117.9 (B) 132.3 (C) 137.9 (D) 157.9

Answer (D)

Solution.

Hoek- Brown failure criteria by :-

$$\sigma_1 = \sigma_3 + \sigma_{ci} \left(m \cdot \frac{\sigma_3}{\sigma_{ci}} + s \right)^a$$

Where,

σ_1, σ_3 are the major and minor effective principal stresses at failure

σ_{ci} is uniaxial compressive strength of the intact rock material

m, s and a are material constant

Therefore,

$$\sigma_1 = 20 + 80 \left(7.88 \times \frac{20}{80} + 1 \right)^{0.5}$$

$$\sigma_1 = 157.9 \text{ MPa}$$

Q.47 A wire of length L is cut into two pieces to construct a circle and an equilateral triangle such that the combined area is minimum. The length of the wire used to construct the circle is

(A) $\frac{\sqrt{3}\pi L}{9+\sqrt{3}\pi}$

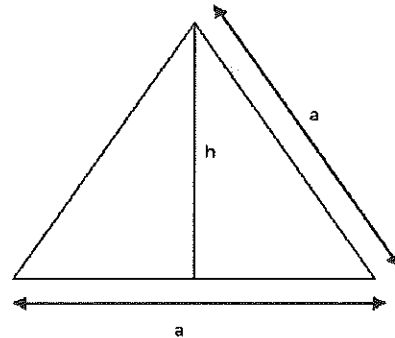
(B) $\frac{9L}{9+\sqrt{3}\pi}$

(C) $\frac{L}{2}$

(D) $\frac{18L}{9-\sqrt{3}\pi}$

Answer (A)

Solution.



$$h = \frac{a\sqrt{3}}{2}$$

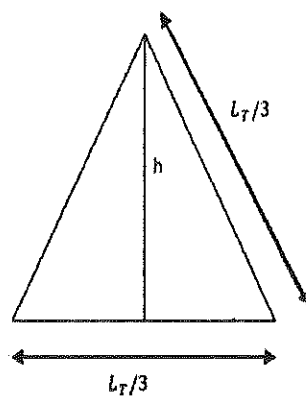
Area = $\frac{1}{2} \times \text{base} \times \text{height}$

Let total length of wire used to construct equilateral triangle and circle = L

Then, triangle length = L_T

And circle length = $L - L_T$

Now, calculation of equilateral triangle area –



$$h = \frac{L_T \sqrt{3}}{6}$$

$$\text{base} = \frac{L_T}{3}$$

$$\therefore A_T = \frac{1}{2} \times \frac{L_T}{3} \times \frac{L_T \sqrt{3}}{6}$$

$$\therefore A_T = \frac{L_T^2}{12\sqrt{3}}$$

Now, calculation of circle area –

$$\text{Circumference} = 2\pi r$$

$$\therefore L - L_T = 2\pi r$$

$$\text{Or } r = \frac{L - L_T}{2\pi}$$

$$\therefore \text{Area of circle } (A_c) = \pi r^2$$

$$A_c = \frac{\pi (L - L_T)^2}{4\pi^2}$$

$$A_c = \frac{(L - L_T)^2}{4\pi}$$

Now, combined area –

$$A = \frac{L_T^2}{12\sqrt{3}} + \frac{(L - L_T)^2}{4\pi}$$

$$\frac{dA}{dL_T} = \frac{L_T}{6\sqrt{3}} - \frac{L - L_T}{2\pi}$$

For max. or min.

$$\frac{dA}{dL_T} = 0$$

$$\frac{L_T}{6\sqrt{3}} - \frac{L - L_T}{2\pi} = 0$$

$$\therefore L_T = \frac{3\sqrt{3}L}{\pi + 3\sqrt{3}}$$

Now, length of the circle = $L - L_T$

$$= L - \frac{3\sqrt{3}L}{\pi + 3\sqrt{3}} \dots\dots\dots(1)$$

Solving equation (1), length (L_c) of the wire used to construct circle is :-

$$L_c = \frac{\sqrt{3} \cdot \pi \cdot L}{9 + \sqrt{3} \cdot \pi}$$

Common Data for Questions 48 and 49:

Pressure characteristic of a mine fan is given by $P = -0.06Q^2 + 400$, where P is the pressure in Pa and Q^2 is the quantity in m^3/s . The resistance of the mine is $0.19 \text{ N s}^2/m^8$.

Q.48 The mine quantity in m^3

- (A) 160.0 (B) 53.5 (C) 45.9 (D) 40.0

Answer (D)

Solution.

We have,

$$P = RQ^2$$

$$-0.06Q^2 + 400 = 0.19 \times Q^2$$

$$\therefore Q = \sqrt{\frac{400}{0.25}} = 40 \text{ m}^3/\text{sec}$$

Q.49 An identical fan is installed in the mine to operate in series with the existing fan.

The new mine quantity in m^3/s is

- (A) 75.6 (B) 56.7 (C) 50.8 (D) 30.2

Answer (C)

Solution.

When identical fan is installed in series with the existing fan, then new pressure becomes

$$P' = 2P = -0.06 \times 2 \times Q^2 + 2 \times 400$$

$$P' = -0.12Q^2 + 800$$

$$\text{Now, } P' = RQ^2$$

$$-0.12Q^2 + 800 = 0.19Q^2$$

$$\therefore Q = \sqrt{\frac{800}{0.31}} = 50.8 \text{ m}^3/\text{sec}$$

Common Data for Questions 50 and 51:

The following observations are taken during a closed traverse.

Side	Length (m)	WCB
AB	100	90°
BC	173	180°
CA	200	330°

Q.50 The closing error of the traverse in mm is

- (A) 205 (B) 20.5 (C) 2.05 (D) 0.205

Answer (A)

Solution.

Sum of the latitudes of the lines :-

$$\sum L \cos \theta = 100 \cos 90^\circ + 173 \cos 180^\circ + 200 \cos 330^\circ$$

$$\sum L \cos \theta = 0.2050 \text{ m}$$

Sum of the longitudes of the lines :-

$$\sum L \sin \theta = 100 \sin 90^\circ + 173 \sin 180^\circ + 200 \sin 330^\circ$$

$$\sum L \sin \theta = 0$$

Now,

$$\begin{aligned}
 \text{Closing error of traverse} &= \sqrt{(\sum L \cos \theta)^2 + (\sum L \sin \theta)^2} \\
 &= \sqrt{(0.2050)^2 + (0)^2} \\
 &= 0.2050 \text{ m} \\
 &= 205 \text{ mm}
 \end{aligned}$$

- Q.51 The reduced bearing of the closing error in degrees is
- (A) 87.21 (B) 64.03 (C) 14.04 (D) 0

Answer (D)

Solution.

$$\text{Reduced bearing of closing error} = \tan^{-1} \left(\frac{\sum L \sin \theta}{\sum L \cos \theta} \right)$$

$$\begin{aligned}
 \therefore \text{Reduced bearing of closing error} &= \tan^{-1} \left(\frac{0}{0.205} \right) \\
 &= 0^\circ
 \end{aligned}$$

Statement for Linked Answer Questions 52 and 53:

Economic analysis of an iron ore deposit reveals that the net value of the ore is related to the grade mined as shown below:

Grade (%Fe)	Net value of ore (Rs/tonne)
64.5	3200
60.2	1800

- Q.52 Assuming linear relationship between the net value and grade, the break-even cut-off grade in % Fe is

(A) 52.2

(B) 54.7

(C) 58.0

(D) 62.2

Answer (B)

Solution.

Assuming the net value (Rs/tonne) and grade (% Fe) relationship is linear then an equation of form $y = a + bx$ relating to net value (y) to grade (x)

So,

$$3200 = a + 64.5 b \dots\dots\dots(1)$$

$$1800 = a + 60.2b \dots\dots\dots(2)$$

Solving both equations,

$$a = -17800$$

$$b = 325.58$$

Therefore, result is

$$Y = -17800 + 325.58 x$$

But , at break even cutoff grade net value (Y) is zero, i.e

$$Y = 0$$

$$\therefore 0 = -17800 + 325.58 x$$

$$\therefore x = \frac{17800}{325.58} = 54.7\%$$

Q.53 Assuming that the grade follows normal distribution with mean 62.7%, and standard deviation 10.0% (A portion of the standard normal distribution table is given below),

z	0.00	0.01	0.02	0.03	0.04
0.6	0.72575	0.72907	0.73237	0.73565	0.73891
0.7	0.75803	0.76115	0.76424	0.76730	0.77035
0.8	0.78814	0.79103	0.79389	0.79673	0.79954
0.9	0.81594	0.81859	0.82121	0.82381	0.82639
1.0	0.84134	0.84375	0.84613	0.84849	0.85083

the percentage of waste in the deposit based on the break-even cut-off grade is

(A) 78.8

(B) 71.2

(C) 28.8

(D) 21.2

Answer (D)

Solution.

From normal distribution, we have

$$z = \frac{x - \mu}{\sigma}$$

$$\because x = 54.7\% \quad \mu = 62.7\%, \sigma = 10\%$$

$$z = \frac{54.7 - 62.7}{10}$$

$$z = \frac{-8}{10} = -0.8$$

From normal distribution table, at $z = -0.8$

The percentage of ore in deposit based on cut off grade is = 78.814 %

Therefore,

The percentage of waste in deposit based on the cut off grade is = $100 - 78.814$

$$= 21.2\%$$

Statement for Linked Answer Questions 54 and 55:

A 4.6 m wide vein dipping at 80° is mined by horizontal cut-and-fill stoping method. The fill is to be placed in the stope along the length of 46 m and to a height of 3.0 m. If the specific weight of the fill material is 15.86 kN/m^3 and the

porosity is 35%, under fully saturated conditions

Q.54 The volume of water in the fill in m^3 is

- (A) 222.18 (B) 332.40 (C) 336.44 (D) 634.80

Answer (A)

Solution.

$$\text{Porosity} = \frac{\text{volume of void (air/water)}}{\text{total volume of rock sample}}$$

$$\frac{35}{100} = \frac{\text{volume of water}}{4.6 \times 46 \times 3}$$

$$\therefore \text{volume of water} = \frac{35 \times 4.6 \times 46 \times 3}{100}$$

$$\therefore \text{volume of water} = 222.18 \, m^3$$

Q.55 The mass of solids in saturated fill in tonnes is

- (A) 820.00 (B) 804.10 (C) 799.30 (D) 788.80

Answer (B)

Solution.

$$\text{Specific weight} = \frac{\text{weight of solid}}{\text{volume of solid}}$$

$$15.86 = \frac{\text{weight of solid}}{4.6 \times 46 \times 3}$$

$$\therefore \text{weight of solid} = 10076.928 \, \text{kN}$$

$$= 10076928 \, \text{N}$$

$$\text{mass of solid} = 1026292.355 \, \text{kg}$$

$$\text{And mass of water} = 222.18 \times 1000 = 222180 \, \text{kg}$$

$$\text{Hence mass of solids in saturated fill} = 1026292.355 - 222180$$

$$= 804112 \, \text{kg}$$

= 804.1 tonnes

Q.56 If $3 \leq X \leq 5$ and $8 \leq X \leq 11$ then which of the following options is TRUE?

(A) $\frac{3}{5} \leq \frac{X}{Y} \leq \frac{8}{5}$

(B) $\frac{3}{11} \leq \frac{X}{Y} \leq \frac{5}{8}$

(C) $\frac{3}{11} \leq \frac{X}{Y} \leq \frac{8}{5}$

(D) $\frac{3}{5} \leq \frac{X}{Y} \leq \frac{8}{11}$

Answer (B)

Solution.

Putting $X = 4$ and $Y = 9$, it is found that option (B) follows nearest values and correct order as compared to other given option and hence option (B) is correct.

Q.57 The Headmaster _____ to speak to you.

Which of the following options is incorrect to complete the above sentence?

(A) is wanting (B) wants (C) want (D) was wanting

Answer (C)

Q.58 Mahatama Gandhi was known for his humility as

(A) he played an important role in humiliating exit of British from India.

(B) he worked for humanitarian causes.

(C) he displayed modesty in his interactions.

(D) he was a fine human being.

Answer (C)

Q.59 All engineering student should learn mechanics, mathematics and how to do

I II III IV

computation .

Which of the above underlined parts of the sentence is not appropriate?

- (A) I (B) II (C) III (D) IV

Answer (D)

Q.60 Select the pair that best expresses a relationship similar to that expressed in the pair:

water: pipe::

- (A) cart: road (B) electricity: wire
- (C) sea: beach (D) music: instrument

Answer (B)

Q.61 Velocity of an object fired directly in upward direction is given by $V = 80 - 32t$, where t is time in seconds. When will the velocity be between 32 m/sec and 64 m/sec?

- (A) $(1, 3/2)$ (B) $(1/2, 1)$ (C) $(1/2, 3/2)$ (D) $(1, 3)$

Answer (C)

Solution.

$$V = 80 - 32t$$

When $V = 32$ m/sec

$$32 = 80 - 32t$$

$$\therefore t = 3/2 \text{ sec}$$

When $V = 64$ m/sec

$$64 = 80 - 32t$$

$$\therefore t = 1/2 \text{ sec}$$

$$\therefore (1/2, 3/2)$$

Q.62 In a factory, two machines M1 and M2 manufacture 60% and 40% of the autocomponents respectively. Out of the total production, 2% of M1 and 3% of M2 are found to be defective. If a randomly drawn autocomponent from the combined lot is found defective, what is the probability that it was manufactured by M2?

(A) 0.35

(B) 0.45

(C) 0.5

(D) 0.4

Answer (C)

Solution.

For machine M_1 , given that

$$P(B) = 60\% = 0.60$$

$$P(A/B) = 2\% = 0.02$$

$$\therefore P(B) \cdot P(A/B) = 0.012$$

For machine M_2 , given that

$$P(B) = 40\% = 0.40$$

$$P(A/B) = 3\% = 0.03$$

$$\therefore P(B). P(A/B) = 0.012$$

$$\text{And } \sum P(B). P(A/B) = 0.024$$

Hence , required probability by M_2 is given by:-

$$P(M_2) = \frac{\{P(B).P(A/B)\}_{M_2}}{\sum P(B).P(A/B)}$$

$$= \frac{0.012}{0.024} = 0.5$$

Q.63 **Following table gives data on tourists from different countries visiting India in the year 2011.**

Country	Number of tourists
USA	2000
England	3500
Germany	1200
Italy	1100
Japan	2400
Australia	2300
France	1000

Which two countries contributed to the one third of the total number of tourists who visited India in 2011?

(A) USA and Japan

(B) USA and Australia

(C) England and France

(D) Japan and Australia

Answer (C)

Solution.

Total number of tourists who visited India in 2011 = 13500

Now, one third of total number of tourists = 4500

Total number of tourists from England and France = 4500

Hence, England and France both countries contributed to the one third of the total number of tourists who visited India in 2011.

Q.64 If $|-2X + 9| = 3$ then the possible value of $|-X| - X^2$ would be:

(A) 30

(B) -30

(C) -42

(D) 42

Answer (B)

Solution.

$$|-2X + 9| = 3$$

$$\text{Or } -2X + 9 = \pm 3$$

Taking (+)

$$\text{Then } X = 3$$

Therefore ,

$$|-X| - X^2 = 3 - 9 = -6, \text{ and}$$

Taking (-)

$$\text{Then } X = 6$$

Therefore ,

$$|-X| - X^2 = 6 - 36 = -30$$

$\therefore$ possible value = -30

Q.65 All professors are researchers

Some scientists are professors

Which of the given conclusions is logically valid and is inferred from the above arguments:

- | | |
|-------------------------------------|-----------------------------------|
| (A) All scientists are researchers | (B) All professors are scientists |
| (C) Some researchers are scientists | (D) No conclusion follows |

Answer (C)